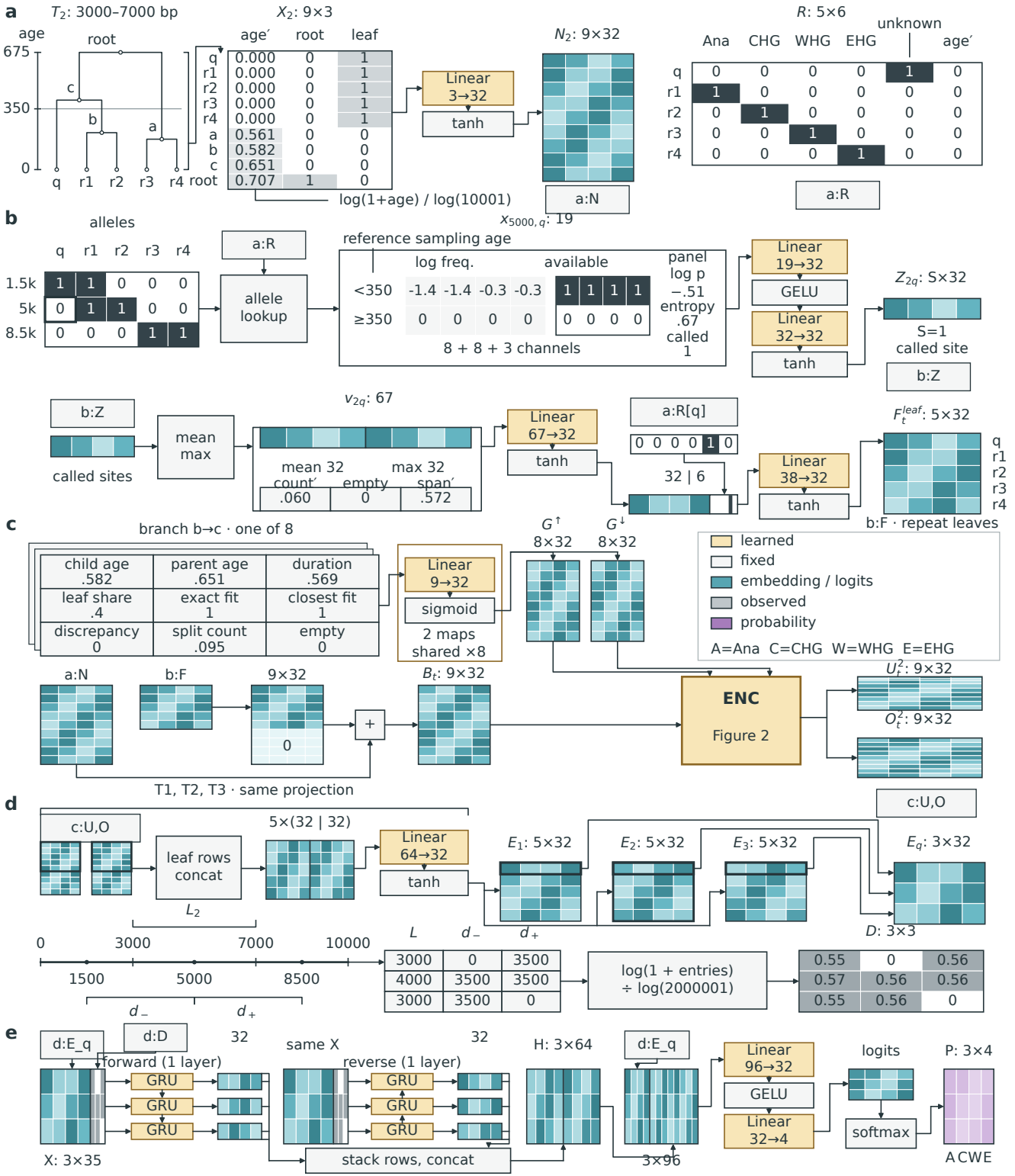

a
$T_2$: 3000–7000 bp
$X_2$: 9×3
$N_2$: 9×32
$R$: 5×6
unknown
Ana CHG WHG EHG age´
root leaf
Linear 3→32
tanh
a:N
a:R
log(1+age) / log(10001)
$x_{5000,q}$: 19
b
alleles
reference sampling age
log freq. available
panel log p −.51 entropy .67 called 1
8 + 8 + 3 channels
allele lookup
Linear 19→32
GELU
Linear 32→32
tanh
$Z_{2q}$: S×32
S=1 called site
b:Z
called sites
mean max
$v_{2q}$: 67
mean 32 count´ .060
empty 0
max 32 span´ .572
Linear 67→32
a:R[q]
32 | 6
Linear 38→32
$F_t^{leaf}$: 5×32
b:F · repeat leaves
c
branch b→c · one of 8
child age .582
parent age .651
duration .569
leaf share .4
exact fit 1
closest fit 1
discrepancy 0
split count .095
empty 0
Linear 9→32
sigmoid
2 maps shared ×8
$G^{\uparrow}$ 8×32
$G^{\downarrow}$ 8×32
learned
fixed
embedding / logits
observed
probability
A=Ana C=CHG W=WHG E=EHG
a:N
b:F
9×32
$B_t$: 9×32
ENC
Figure 2
$U_t^2$: 9×32
$O_t^2$: 9×32
T1, T2, T3 · same projection
d
c:U,O
5×(32 | 32)
leaf rows concat
$L_2$
Linear 64→32
tanh
$E_1$: 5×32
$E_2$: 5×32
$E_3$: 5×32
$E_q$: 3×32
$D$: 3×3
0 3000 7000 10000
1500 5000 8500
$d_-$ $d_+$
L $d_-$ $d_+$
3000 0 3500
4000 3500 3500
3000 3500 0
log(1 + entries) ÷ log(2000001)
0.55 0 0.56
0.57 0.56 0.56
0.55 0.56 0
e
d:E_q
d:D
32
same X
32
forward (1 layer)
reverse (1 layer)
GRU
H: 3×64
d:E_q
Linear 96→32
GELU
Linear 32→4
logits
softmax
P: 3×4
X: 3×35
stack rows, concat
3×96
A C W E